

Ansys Long-Term Technology Strategy 2021



NUMERICAL METHODS

- Solver methods
- IGA/XFEM/Inverse
- Stochastic & Bayesian Hybrid Models



AI/MACHINE LEARNING

- Analysis Productivity
- Augmented Simulation
- Data Driven, Physics Informed, ML Based



DIGITAL TWINS

- End to end solution architecture
- Data analytics/AI-ML/Hybrid
- Reduced Order Modeling



MESHING / GEOMETRY

- Non- & Conformal Meshing
- Morphing, Immerse-Boundary
- Parallel Meshing



HIGH PERFORMANCE COMPUTING

- Task, Domain, Thread based
- Fine grain (GPU)
- Exascale & Quantum computing



ICME and Additive

- Materials Intelligence & Selection
- ICME & Multiscale Modeling
- Additive Science



VISUALIZATION AND UX/UI

- Augmented Reality/Virtual Reality
- Immersive User Experiences
- Safety Critical HMI



CLOUD

- Cloud First
- On-Prem, Private, Public
- HPC as a Service



SOLUTIONS

- Autonomy
- Electrification
- 5G



PLATFORMS/WORKFLOWS

- Multiphysics, Multi-domain
- Process Integration, & Optimization
- Simulation Process & Data Management



MODEL BASED SYSTEM ENGINEERING

- Collaborative Architectural Modeling
- Virtual Verification and Validation
- Lifecycle Trade Analysis & Optimization



NEW VERTICAL HEALTHCARE

- Biopharma drug discovery
- Medical devices & equipment
- Clinical apps



The world around us is governed by the laws of physics, which are modeled by second-order partial differential equations, such as the Navier-Stokes equations for fluids or Maxwell's equations for electromagnetics. We simulate the physics numerically using methods such as finite element analysis (FEA) used in our Ansys Mechanical solver or finite volume methods used in our Ansys Fluent fluid dynamics solver. These numerical solvers use well-known mathematical principles on well-known shapes – such as squares or triangles – that are overlaid as a mesh on a model of what's being simulated. The mesh is created by the simulation software, which uses it to calculate how the individual cells will react to outside stresses like heat, pressure or electromagnetic interference, and how they will interact with one another. When combined in a simulation run, those reactions and interactions of cells in the mesh enable the simulation software to predict whether that falling soda can will burst on impact, how to minimize electromagnetic interference, how to best cool a PCB to prevent cracking, or just about anything else you can imagine.

Simulation is a tested means of improving existing products because

it enables engineers to explore “what-if” questions without going to the time and expense of creating a physical prototype for every scenario. Simulation is also used to help engineers understand the best way to design something completely new.

Simulating impacts, electromagnetic interference, fluid flows and more is common today, as is relying on simulation to suggest optimized design alternatives. However, simulations can get much more complex. They can take into account the effects of multiple physics simultaneously over distinct domains, or in large assemblies and systems, or when scaling from the microscopic to the mission level. These complex simulations require significant compute power to solve in a reasonable time if maximum accuracy is required.

Humans aren't immune to the effects of ballooning product design and development complexity, either. Knowing how to set up, run and modify advanced simulations requires significant expertise. The more complex the simulation, the more expertise is required. This leaves a relatively small pool of simulation experts, which impedes the ubiquity of simulation.

Simulation Without Compromise

Technologies already under development promise to simplify and democratize the use of simulation by allowing more people to solve more multiphysics, multi-domain, multi-scale problems with:

1. High speed
2. High accuracy
3. Easy-to-use workflows
4. Robustness to ensure repeatable results

Unfortunately, there are tradeoffs involved with these four pillars. For a given problem, you can make the simulation run faster by making a mesh with fewer points, but then you lose accuracy. You can have higher accuracy with 10X more mesh points, but then the simulation runtimes increase significantly.

Using HPC and AI/ML, Ansys has made our simulation tools run more quickly, more accurately and more robustly while making them easier to use. Still, many of the tools need hundreds of parameters to be set by an advanced engineer so that the solver runs in the best possible manner. In the future, we will have AI/ML automatically learn to set these parameters. We can imagine a future state where a designer will give voice instructions using natural